

Hivemind

Three control-plane attacks

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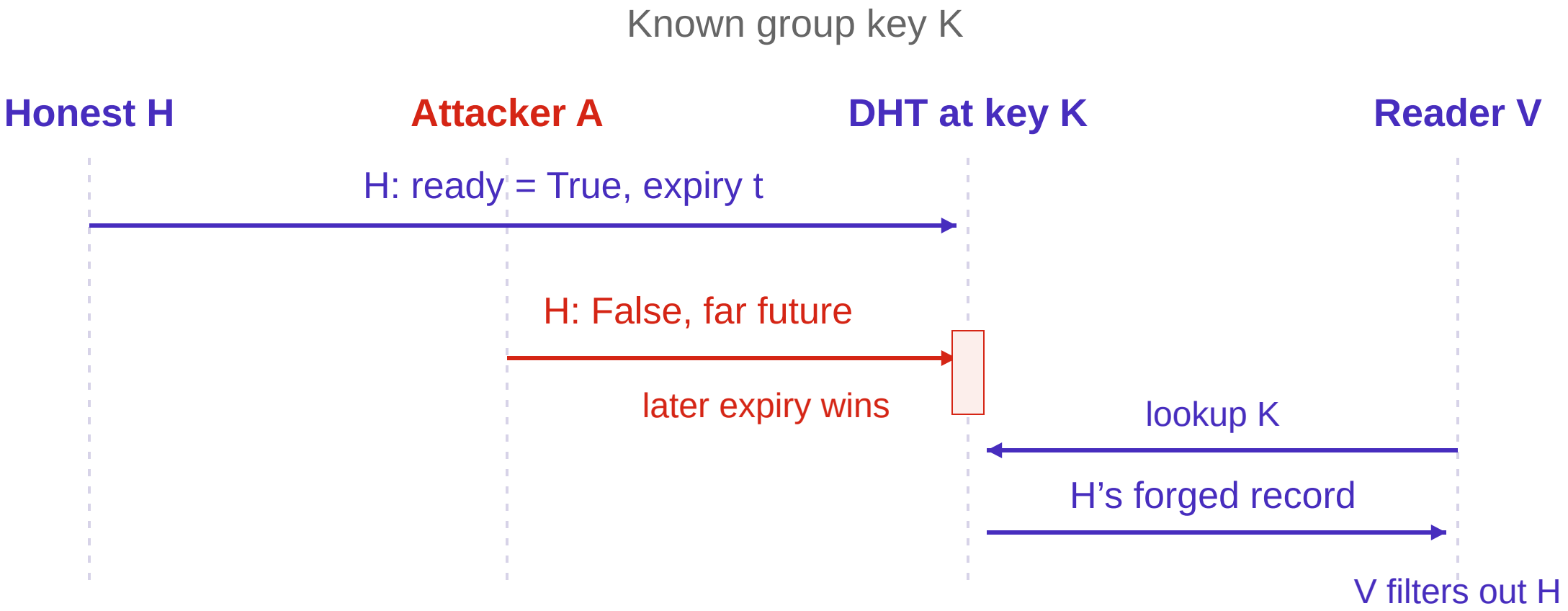
Secura Lab

Chalmers University of Technology | 2 October 2026

Last week → today

Last week's questions	Current status
Can we force leadership without knowing the group key K?	Ongoing
Crawl a live swarm every d seconds and record who has advertised.	No live averaging swarm found. Need a known group key to crawl.
Measure lookup time and the shortest advertised deadline.	Local tests using RIPE Atlas ping RTTs: Sweden, US, Brazil, India, Australia.

1. Forged withdrawal



H stays online, but disappears from group discovery.

H’s refresh expires sooner; the DHT keeps the record that expires later.

Withdrawal: two separate tests

1 Can an attacker hide honest peers?

12 or 40 honest advertisers, plus one attacker.

Check whether forged withdrawals hide peers, even after they advertise again.

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2 Does hiding a peer make the target join the attacker?

Use RIPE Atlas delays in a local swarm.

Compare who the target joins, with and without a forged withdrawal.

Withdrawal: results

Check	Observed result
Targeted peers hidden	39/39 (40 honest peers)
Refreshes that restored the peer	0/30 attempts
Target joined attacker, with withdrawal	0/10 trials
Target joined attacker, without withdrawal	0/10 trials

Group capture was tested locally with delays derived from RIPE Atlas pings.

Exact tables → A1

2. Deadline inflation

Keep H visible, but move its deadline behind the attacker's.

Before

Peer	Advertised deadline
H	now + 5 s
A	now + 6 s

V would contact H

forge H's expiration

After forged write

Peer	Advertised deadline
A	now + 6 s
H	now + 1 hour

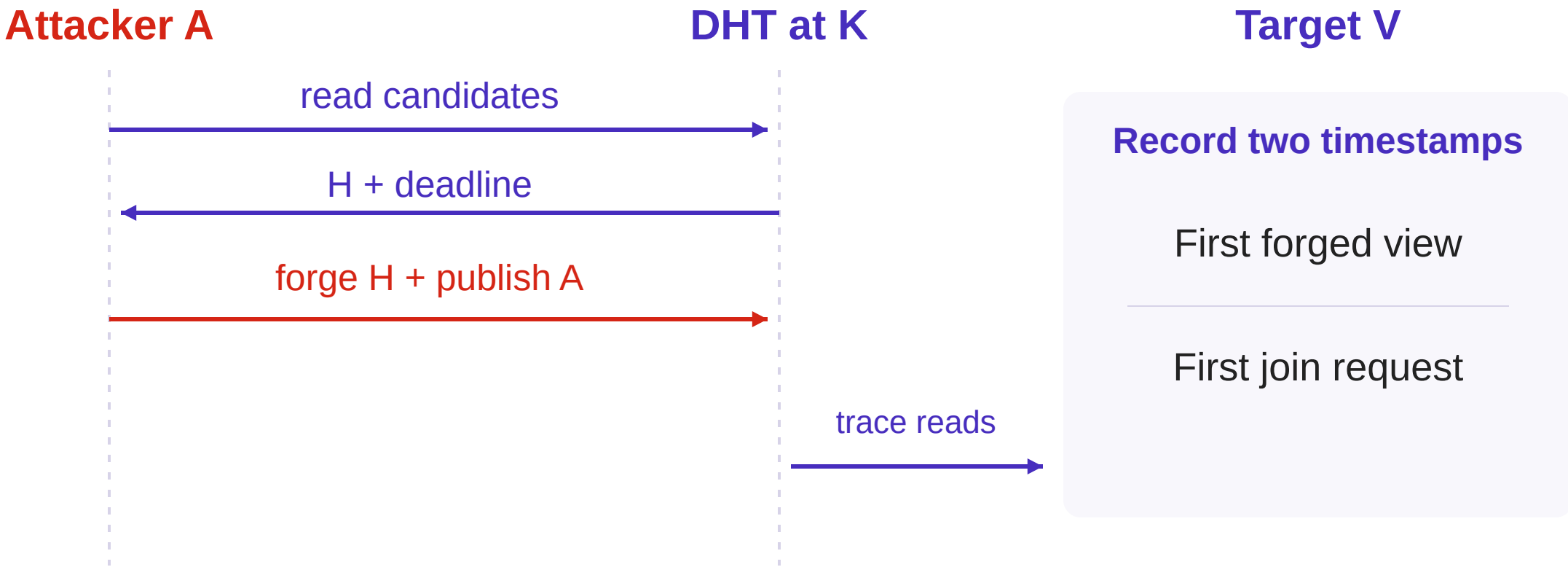
V may contact A

Only if V reads the changed record before choosing whom to join.

Illustrative times · H's ready flag stays True · no crafted DHT identity

Inflation: measure the race to V’s view

A’s own deadline stays later than H’s; only H’s record is forged.



Compare timestamps: forged view before first join request?

28 local DHT nodes · 5 RTT origins · 5–7 s / 11–15 s windows · paired control

Accepted writes did not change V’s join

Measured event	Attack trials
DHT accepted the forged record	**10 / 10**
Forged view recorded before V’s first request	**0 / 10**
V joined the attacker	**0 / 10**

No-forgery control: 0 / 10 targets joined A.

In 7/10 trials: V picked a peer → forged view appeared later

One reactive, single-peer policy; other strategies remain untested.

Exact tables → A2

3. One group, conflicting reducer indices

Precondition: A is already the group leader.



different
ordered rosters

Same group ID
Same four peers

Reducer index

	0	1	2	3
A's own view	A	F1	F2	F3
Sent to F1	F1	A	F2	F3
Sent to F2	F2	F1	A	F3
Sent to F3	F3	F1	F2	A

Each recipient claims A's reducer slot (index 0 in this example).

Equal-sized slices can be mixed; followers do not cross-check one common roster.

Roster test: change only the ordering

1 Same inputs

A	0	1	2	4
F1	0	1	2	4
F2	0	1	2	4
F3	0	1	2	4

Four real local averagers

2 Two conditions

One common roster

5 paired trials

Different roster per peer

3 Check outputs

Expected average

0	1	2	4

Did it complete?

Is the tensor correct?

Were failures recorded?

Real DHT + join streams + stock all-reduce · leadership arranged in advance

Wrong tensors, successful steps

One recorded trial

Correct average

0	1	2	4

One attacked peer’s output

1.75	0.75	1.5	3.0

Across five paired trials

Corrupted trials	
Control	**0 / 5**
Attack	**5 / 5**

20/20 attacked outputs wrong

All 40 steps completed; tested failure indicators stayed clear.

Tensor corruption measured locally; downstream training damage was not tested.

Exact tables → A3

4. Deadline undercutting

Read every advertised deadline, then declare one step earlier than all of them.

Advertised at K

Peer	Deadline
H1	now + 11.4 s
H2	now + 12.8 s
H3	now + 13.8 s
A	now + 11.4 s − 1 ulp

undercut the earliest →

Leader election

earliest deadline wins

A leads; A admits whom it likes

Cost to A: one identity,
no compute, no crafted node ID

One ulp is 2.4×10^{-7} s here · enough to win the comparison, no advertisement life lost

Undercutting: two experiments

Same key, node IDs and honest schedule; only the attacker's region changes.

Read ads at K → **Undercut earliest** → **Publish own ad** → **Wait for joins**

Experiment 1 · where the attacker sits

5 origins: Stockholm, New York, São Paulo, Mumbai, Sydney · RTTs from RIPE Atlas pings

2 honest windows: 5–7 s and 11–15 s

3 seeds each → 30 runs

Experiment 2 · is the failure ours?

Our attacker skips any ad with under 3.5 s left (request_timeout 2.5 + our own 1.0 margin)

Set both that margin and the 0.3 s confirmation wait to zero · Stockholm, the cheapest origin

28 local DHT nodes, 5–6 per region · above bucket_size 20, so lookups must route

Injected delays on one machine · no loss, churn or connection setup · key known to A

The delay profile: RIPE Atlas anchor RTTs

RIPE Atlas is a global measurement network whose anchor hosts continuously ping each other; we took the median of three ICMP replies per directed pair.

from \ to	SE	US	BR	IN	AU
Stockholm	9	99	209	147	298
New York	99	13	125	285	240
São Paulo	209	111	16	228	305
Mumbai	147	240	228	23	468
Sydney	299	239	305	503	12

233 ms

median between regions

99 – 503 ms

range across the 20 links

Each region holds 5–6 of the 28 nodes in the swarm

Link RTTs, not DHT operation times · measured 30 Sep 2026 · profile saved as ripe_anchor_rtt_20260930.json

The window decides, not the attacker

Attacker origin	11–15 s window	5–7 s window
Stockholm	**3 / 3**	**0 / 3**
New York	**3 / 3**	**0 / 3**
São Paulo	**3 / 3**	**0 / 3**
Mumbai	**3 / 3**	**0 / 3**
Sydney	**3 / 3**	**0 / 3**

Location changes cost, not outcome: read + write 2.1 s from Stockholm, 3.7 s from Sydney

Removing our own caution: published 1/3 → 3/3, led 0/3 → 1/3, captured 0/3 either way

Figure: results/step19_region_sweep.png · raw runs: step19_wan_delay_*_r5_*.csv

A1. Withdrawal discovery tests

One ordinary random-ID writer · counts below matched in both repeats

Honest peers	Targets / run	Hidden / run	Others visible	Repeats
12	1	1	11 / 11	2
12	3	3	9 / 9	2
12	6	6	6 / 6	2
12	11	11	1 / 1	2
40	1	1	39 / 39	2
40	20	20	20 / 20	2
40	39	39	1 / 1	2

Separate refresh test: 0/30 attempts restored H in each swarm size.

Source: step8_withdrawal_phase2.csv and step8_withdrawal_phase2_n40.csv

Exact tables → A2

A2. Forged-record trials

Targets that joined A / trials · one seed per region and deadline window

RTT origin	Window	Control	Withdrawal	Inflation
Sweden	5-7 s	0 / 1	0 / 1	0 / 1
United States	5-7 s	0 / 1	0 / 1	0 / 1
Brazil	5-7 s	0 / 1	0 / 1	0 / 1
India	5-7 s	0 / 1	0 / 1	0 / 1
Australia	5-7 s	0 / 1	0 / 1	0 / 1
Sweden	11-15 s	0 / 1	0 / 1	0 / 1
United States	11-15 s	0 / 1	0 / 1	0 / 1
Brazil	11-15 s	0 / 1	0 / 1	0 / 1
India	11-15 s	0 / 1	0 / 1	0 / 1
Australia	11-15 s	0 / 1	0 / 1	0 / 1

Forgery accepted in 20/20 attack trials (29/29 writes); all 90 honest steps completed.

28-node local RTT emulation · source: step20 sweep_20261002_a analysis + event CSVs

A3. Reducer-index trials

Four peers · five paired local trials · identical input [0, 1, 2, 4]

Trial	Control correct	Attack correct	Max. abs. error	Steps
1	4 / 4	0 / 4	3.00	8 / 8
2	4 / 4	0 / 4	3.00	8 / 8
3	4 / 4	0 / 4	3.00	8 / 8
4	4 / 4	0 / 4	2.25	8 / 8
5	4 / 4	0 / 4	3.00	8 / 8

No bans, fallback elements, RPC/all-reduce errors, timeouts, or hard cancellations.

Source: step17_roster_live_rounds_confirm_20260928_confirm_t5_s1717.csv